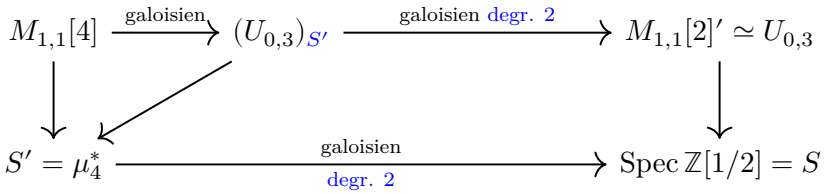
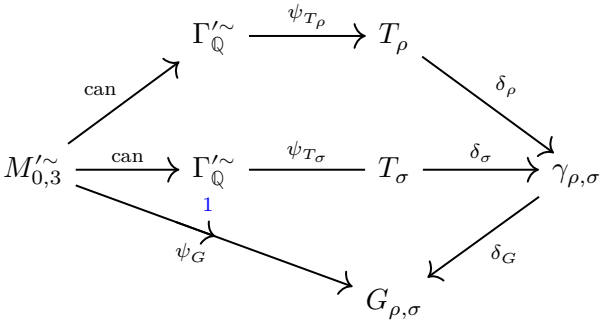
[scan 568]


$M_{1,1}[4]$
galoisien
$(U_{0,3})_{S'}$
galoisien degr. 2
$M_{1,1}[2]' \simeq U_{0,3}$
$S' = \mu_4^*$
galoisien
degr. 2
$\operatorname{Spec} \mathbb{Z}[1/2] = S$

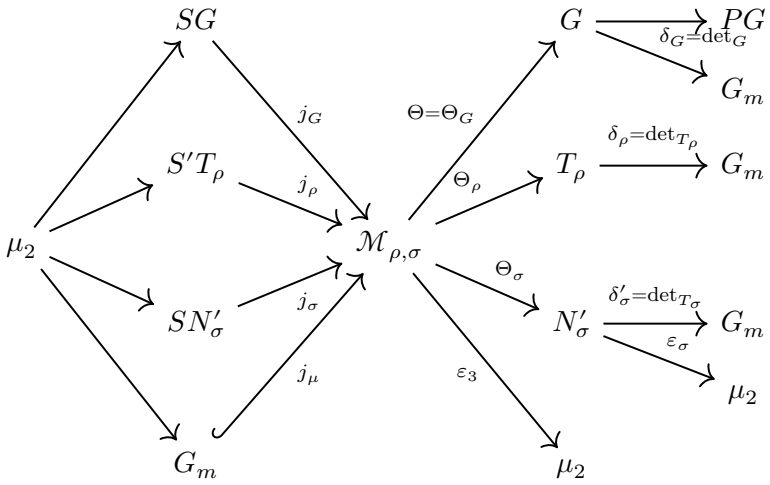
[scan 519]


$\Gamma'^{\sim}_{\mathbb{Q}}$
ψ_{T_ρ}
T_ρ
can
δ_ρ
$M'^{\sim}_{0,3}$
can
$\Gamma'^{\sim}_{\mathbb{Q}}$
ψ_{T_σ}
T_σ
δ_σ
$\gamma_{\rho,\sigma}$
1
ψ_G
δ_G
$G_{\rho,\sigma}$

[scan 536]

$$
\begin{array}{ccccccccc}
1 & \to & L_0 & \to & B_{\rho,\sigma} & \xrightarrow{\delta_B} & \gamma_{\rho,\sigma} & \to & 1 \\
 & & \cup & & \downarrow & & \| & & \\
1 & \to & \mathcal{G} & \to & G_{\rho,\sigma} & \xrightarrow{\delta_G} & \gamma_{\rho,\sigma} & & \\
 & & & & & \searrow^{\chi_G} & \downarrow{\scriptstyle \chi_\gamma} & & \\
 & & & & [& & \widehat{\mathbb{Z}}^* & &]
\end{array}
$$

[scan 448]


SG
$S'T_\rho$
SN'_σ
G_m
μ_2
j_G
j_ρ
j_σ
j_μ
$\mathcal{M}_{\rho,\sigma}$
$\Theta{=}\Theta_G$
Θ_ρ
Θ_σ
ε_3
G
PG
$\delta_G{=}\det_G$
G_m
T_ρ
$\delta_\rho{=}\det_{T_\rho}$
G_m
N'_σ
$\delta'_\sigma{=}\det_{T_\sigma}$
G_m
ε_σ
μ_2
μ_2